

① Hyperbolic Functions

Definitions in exponential form:

$$1) \sinh n = \frac{e^n - e^{-n}}{2}$$

$$2) \cosh n = \frac{e^n + e^{-n}}{2}$$

$$3) \tanh n = \frac{e^n - e^{-n}}{e^n + e^{-n}}$$

$$4) \operatorname{sech} n = \frac{2}{e^n + e^{-n}} = \frac{1}{\cosh n}$$

$$5) \operatorname{cosech} n = \frac{2}{e^n - e^{-n}} = \frac{1}{\sinh n}$$

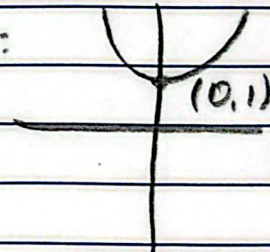
$$6) \coth n = \frac{e^n + e^{-n}}{e^n - e^{-n}} = \frac{1}{\tanh n}$$

Graphs (with asymptotes):

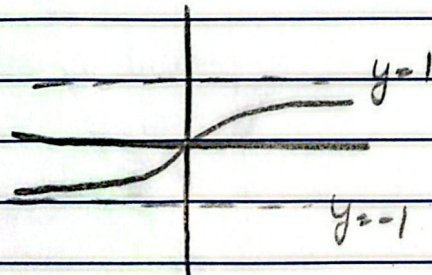
$\sinh n$:



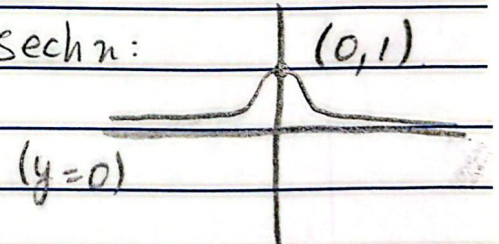
$\cosh n$:



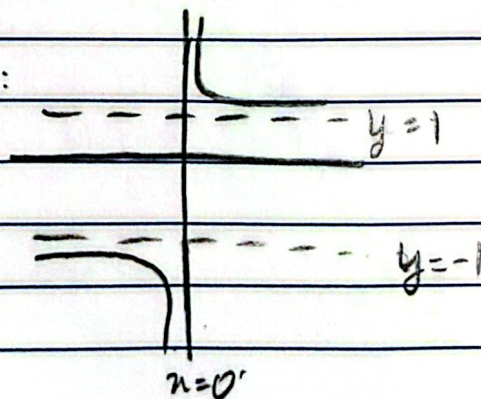
$\tanh n$:



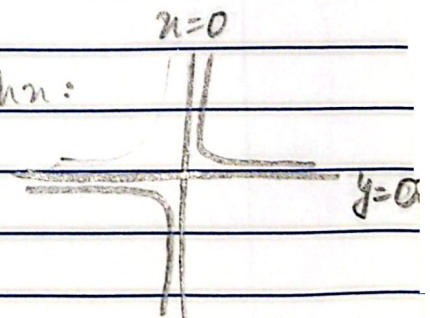
$\operatorname{sech} n$:



$\coth n$:



$\operatorname{cosech} n$:



- To prove hyperbolic identities, use the exponential forms. Ensure proper squaring of certain terms.
- Inverse hyperbolics: (given in MF19)

$$\cosh^{-1} n = y$$

$$\cosh y = n$$

$$\frac{e^y + e^{-y}}{2} = n$$

$$\frac{e^y + 1}{e^y} = 2n$$

$$e^{2y} + 1 = 2ne^y$$

$$e^{2y} - 2ne^y + 1 = 0$$

$$\frac{2n \pm \sqrt{4n^2 - 4}}{2} = n \pm \sqrt{n^2 - 1}$$

$$\therefore n + \sqrt{n^2 - 1} = e^y$$

$$\ln(n + \sqrt{n^2 - 1}) = y = \cosh^{-1} n$$

$$\sinh^{-1} n = y$$

$$\sinh y = n$$

$$\frac{e^y - e^{-y}}{2} = n$$

$$\frac{e^y - 1}{e^y} = 2n$$

$$e^{2y} - 1 - 2ne^y = 0$$

$$\frac{2n \pm \sqrt{4n^2 + 4}}{2}$$

$$= n \pm \sqrt{n^2 + 1}$$

$$\therefore \sinh^{-1} n = \ln(n + \sqrt{n^2 + 1})$$

$$\tanh^{-1} n = y$$

$$\therefore \frac{e^y - e^{-y}}{e^y + e^{-y}} = n$$

$$e^y - e^{-y} = n(e^y + e^{-y})$$

$$e^{2y} - 1 = n(e^{2y} + 1)$$

$$e^{2y} - 1 = ne^{2y} + n$$

$$e^{2y} - ne^{2y} = n + 1$$

$$e^{2y} = \frac{n+1}{1-n}, \therefore \tanh n = \frac{1}{2} \ln \left(\frac{1+n}{1-n} \right)$$

$$\cosh^2 n - \sinh^2 n = 1$$

$$1 - \tanh^2 n = \operatorname{sech}^2 n$$

$$\sinh 2n = 2 \sinh n \cosh n$$

$$\cosh 2n = \cosh^2 n + \sinh^2 n$$

$$1 + \coth^2 n = \operatorname{cosech}^2 n$$